PTBXL: eXtra Large ECG Dataset

A comparative analysis between Statistical and Deep Learning

Supervised by: Dr. Soumick Chatterjee

GROUP 5: Paolini Francesco, Pedrotti Simone, Pennino Mario, Pennone Dario, Pisani Paolo

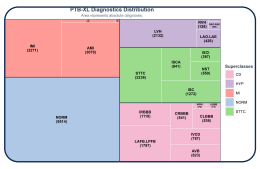
Abstract

Automated ECG classification is currently dominated by Deep Learning (DL) architectures. While powerful, these models often lack the transparency required for clinical decision support. This study evaluates the **performance-interpretability trade-off** by comparing classical statistical models against state-of-the-art DL approaches in a **classification task** (NORM, MI, HYP, STTC, CD) using the **PTB-XL and PTB-XL+ datasets**. In particular, different multi-label approaches have been tested, evaluated and compared with DL benchmarks, as well as Functional Data Analysis applied to the time series.

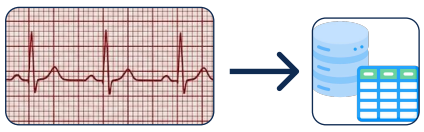
Introduction

DATASET

- **PTB-XL** (Clinical Core): **21,837** 12-lead ECG records (**18,885** patients). Multi-label Taxonomy: 5 Superclasses (NORM, MI, STTC, CD, HYP), 24 Subclasses, plus rhythm/form abnormalities. High multi-morbidity representation.



- **PTB-XL+** (Engineered Features): Raw signals translated into numerical vectors (precise wave durations/amplitudes) via industry-standard algorithms

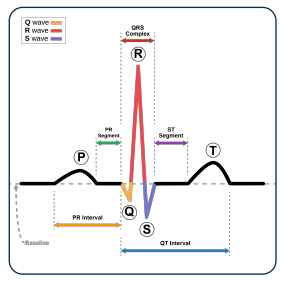


DATA PREPROCESSING

- **Biological Consistency**: Strict physiological bounds applied to filter height/weight/BMI anomalies.
- **Hybrid Imputation**: Missing values resolved via multi-algorithm fusion (12SL + Uni-G) and non-parametric ML (Random Forest via missRanger).
- **Feature Engineering**: Dataset enriched with literature-validated indices (Sokolow-Lyon, Cornell voltage, spatial QRS-T angle) to boost structural defect detection (e.g., HYP).
- **Train/Validation/Test Split**: Dataset featured a pre-defined fold split, allocating 10% each to the validation and test sets; notably, diagnoses in the test fold were physician-validated. Train, validation, and test splits were stratified to maintain the original superclass distribution.

ECG RAW SIGNAL

- **P Wave**: The initial gentle deflection, representing atrial activation (depolarization).
- **QRS Complex**: The primary sharp spike indicating ventricular activation. It consists of the Q wave (initial negative deflection), the R wave (the prominent positive peak), and the S wave (the subsequent negative deflection).
- **T Wave**: The final wave, marking the resting phase (repolarization) of the ventricles.
- **Key Intervals**/Segments:
  - **PR**: Atrioventricular conduction time;
  - **ST**: Ischemia and infarction detection;
  - **QT**: Total ventricular activity time.

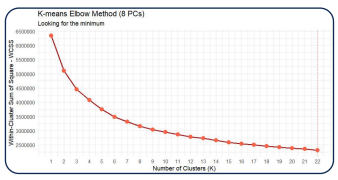


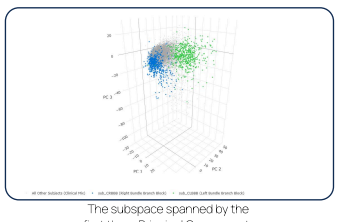
Methodology

STATISTICAL LEARNING

We initially explored the dataset's native geometry using K-Means, K-Medoids, Hierarchical Clustering and DBSCAN in the principal components space.

- **Continous Geometry**: PCA space lacks natural boundaries. K-Means optimization shows no "elbow", and DBSCAN fails by merging almost all data into a single cluster, revealing dense, overlapping multi-morbidities rather than distinct clusters.
- **No Confounding Bias**: Adjusted Rand Index (ARI) metrics are coherent with a clustering driven by ECG morphology, not demographics (age, sex, weight).
- **Unsupervised Limits**: Label Powerset (LBS) suggests that clinical co-morbidities act as continuous topological "bridges", preventing clean geometric separation.
- **PC Interpretation**: Despite the overlap, some structural anomalies subclasses (CLBBB/CRBBB) separate cleanly along PC 2 (ventricular depolarization vector), achieving high Enrichment Factors (EF > 30).





The subspace spanned by the first three Principal Components

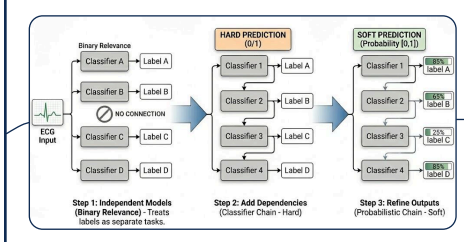
Without natural geometric partitions, supervised architectures are necessary to explicitly force diagnostic boundaries.
Our empirical benchmarking established a clear performance hierarchy across tabular learning architectures:

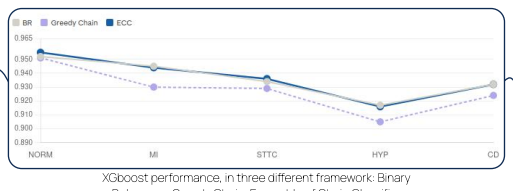
Models

- **Linear Baselines (LDA)**: Underfitting compared to other architectures.
- **Linear Kernelized SVM**: Still a linear method, better performance, but proved computationally heavier than previous models.
- **Bagged Ensembles (Random Forest)**: Reached robust baselines, suggesting that variance-reduction tree is highly suited for ECG data.
- **Gradient Boosting**:
  - **XGBoost**: Non-Linear Method which train sequential trees. The core engine for Classifier and probabilistic chains pipeline.
  - **GPBoost**: Combines XGboost and LMM to evaluate the random effect impact.

To address the complexities of multi-label classification, we implemented a pipeline designed to capture label dependencies:

- **Binary Relevance**: We initiated the pipeline by training independent models for each target class.
- **Classifier Chain**: To overcome the limitations of independence, the output of each model is propagated as an input feature to successive models.
- **Probabilistic Chain**: We transitioned from hard-label inputs to a probabilistic chain, passing predicted class probabilities through the sequence.



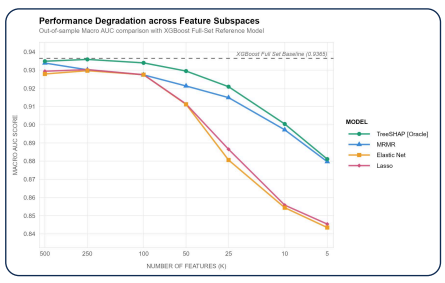


XGboost performance, in three different framework: Binary Relevance, Greedy Chain, Ensemble of Chain Classifier

FEATURE SELECTION

We tried some techniques to reduce the dimensionality of the dataset: using BR XGBoost as the reference model.

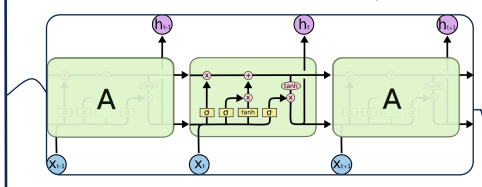
- **mRMR** (Maximum Relevance, Minimum Redundancy): Evaluates global relevance, strictly pairwise.
- **TreeSHAP** (applied on BR XGboost): Exact local attribution, natively handles non-linearity, however it does not address features correlation.
- **Elastic Net** (Using tuned α hyperparameter): Preserves correlated lead groups, but fails due to linear assumptions.
- **LASSO** ($\ell 1$ Regularization): Fails on linearity.

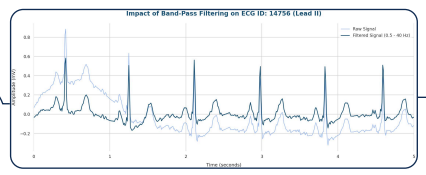


DEEP LEARNING

- **Initial Approach (FFNN on Tabular Data)**
  Preliminary experiments utilizing feed-forward neural networks on the 12sl tabular dataset yielded suboptimal macro and micro AUC scores alongside uninterpretable results, reaffirming the efficacy of traditional statistical learning methods for structured data.

- **Transition to Time-Series Analysis (LSTM)**
  To better leverage deep learning capabilities, a Long Short-Term Memory (LSTM) network was applied to the raw ECG signals (100 Hz sampling rate). Despite implementing a rigorous preprocessing pipeline, utilizing low and high pass filters to attenuate system noise from respiration and muscle artifacts, the resulting performance improvements failed to match established statistical benchmarks.

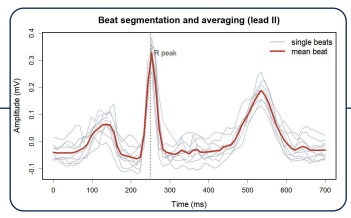


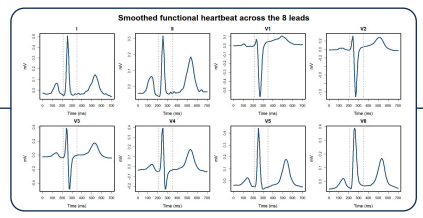


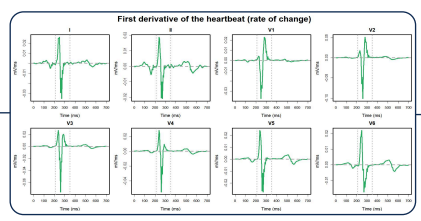
- **Final Architecture (1D-CNN)**
  The application of a 1D Convolutional Neural Network (a fine-tuned ResNet architecture) to the raw ECG data achieved superior results, demonstrating performance metrics comparable to those obtained by traditional statistical methods on the tabular dataset.

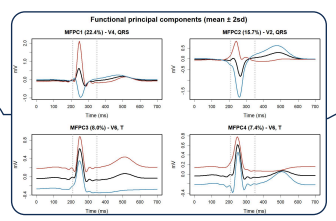
FUNCTIONAL DATA ANALYSIS

- An electrocardiogram is a set of curves. Instead of measuring isolated points, we treat it as **Functional data**: we represent the entire heartbeat as a continuous function across 8 leads and extract its morphology with **Multivariate Functional PCA**. The result feeds models that recognise 5 diagnostic macro-classes.



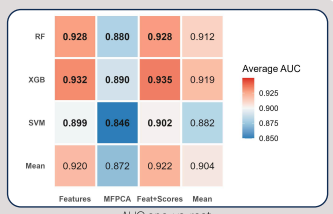






Supervised Learning

- The **XGBOOST** takes the lead over the other algorithms, showing that the 12sl engineered features are more suitable for modeling.
- The **MFPCA** dataset (which is comprised of both 38 MFPCs and 30 derivatives) can compete with much larger datasets.
- Adding the **functional scores** to the engineered features leads to similar performance: the morphological information is largely already captured by the 12SL features.
- The MFPCA's dataset has **slightly bigger AUC** than one created with only the functional scores, showing that the derivatives contribute to morphological explanation.



AUC one-vs-rest

DeLong Test

Derivatives significantly improve detection of myocardial infarction only (**MI: +0.009 AUC, p = 0.001**), consistent with the fact that MI alters the ST–T slope, while adding little elsewhere.

Effect of functional derivatives (DeLong paired test, H0: dAUC = 0)

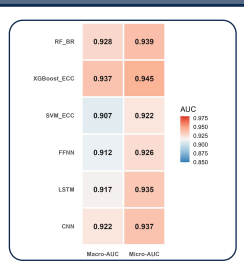
Macroclass	Scores [95% CI]	Scores+deriv. [95% CI]	dAUC	p (DeLong)
NORM	0.912 [0.900-0.924]	0.912 [0.901-0.924]	+0.000	0.8440
MI	0.830 [0.816-0.854]	0.844 [0.826-0.862]	+0.009	0.0011
STTC	0.901 [0.886-0.915]	0.902 [0.888-0.917]	+0.002	0.3480
HYP	0.869 [0.846-0.893]	0.876 [0.851-0.900]	+0.006	0.0614
CD	0.863 [0.843-0.882]	0.867 [0.848-0.886]	+0.004	0.1270

Delong Test scores vs derivates+scores

Results

In this section we present our main results related to the superclasses classification task.

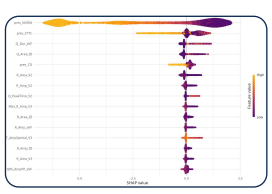
- **Sensitivity, specificity**: Our goal in the choice of the thresholds was to maximise the sensitivity for the the superclasses linked to diseases and specificity for the NORM superclass.



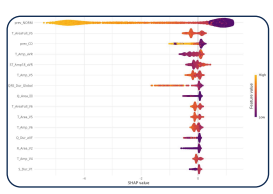


From the Shapley values of our best-performing model, XGBoost, we can further demonstrate the robustness and clinical relevance of our analysis. Indeed, there is a strong correspondence between the importance assigned by the Shapley values to specific features and the medical knowledge underlying each diagnostic superclass:

- **Conduction disturbances** (CD): relevance of PR interval and the QRS complex.
- **Myocardial infarction** (MI) and **ST/T changes** (STTC): closely related from a clinical perspective, as both are associated with ischemic damage, main features related to ST-segment and T-wave morphology, also Q-wave for MI.
- **Hypertrophy** (HYP): again ST-segment abnormalities relevant, but also QRS complex amplitudes.



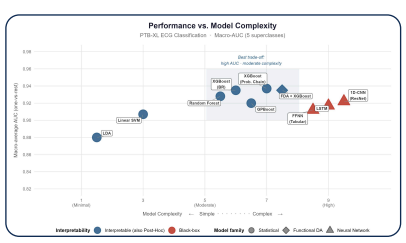
MI BeeSwarm graph, which evaluate the impact of each feature on the prediction of the class



STTC BeeSwarm graph, which evaluate the impact of each feature on the prediction of the class

Conclusion

Our results suggest that statistical learning achieves a performance competitive with DL models on raw signals, while retaining a critical advantage: **clinical interpretability**. Shapley values confirmed strong alignment between feature importance and **medical knowledge**, with PR interval and QRS complex driving CD, ST-segment and T-wave governing MI and STTC, and QRS amplitudes central to HYP detection. Among multi-label strategies, probabilistic chains outperformed Binary Relevance, while greedy approaches suffered from error propagation. Functional Data Analysis further enriched the framework via MFPCA morphological components, and TreeSHAP/mRMR reduced the feature space by **75%** with negligible performance loss — advocating for interpretable, lightweight statistical models as a **robust** alternative to black-box architectures in clinical decision support.



Future Improvements

- Integrate **LMMs** (device) for analyze random effects.
- **Rebalancing**: Try to rescue the rare HYP superclass.
- Scale pipeline to all clinical **subclasses**.
- Apply the pipeline to other engineered features dataset.